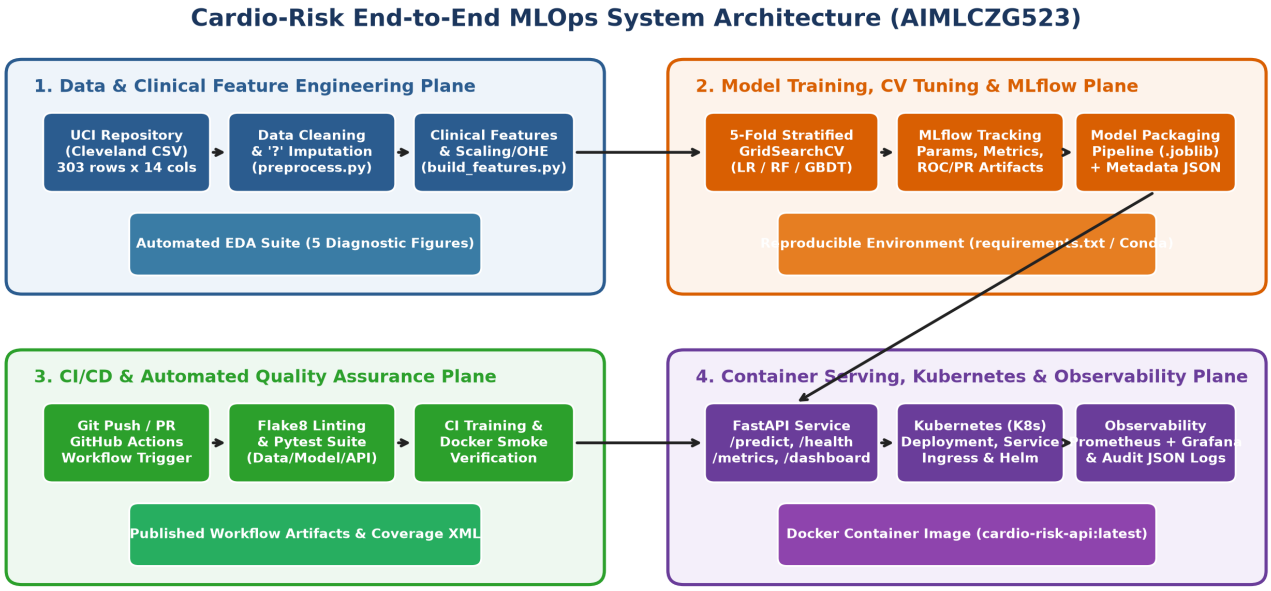
BITS Pilani — Work Integrated Learning Programmes (WILP)

# Machine Learning Operations (MLOps) — AIMLCZG523 | Assignment 01

End-to-End Cardiovascular Risk Classification, MLflow Experiment Tracking, CI/CD Automation, Docker Containerization, Kubernetes Orchestration & Prometheus Monitoring

## 1. Executive Summary & System Architecture

This report presents the complete engineering design, implementation, validation, and production deployment of the **Cardio-Risk MLOps Pipeline** built for the UCI Heart Disease dataset. To eliminate training-serving skew and satisfy production-readiness standards, the system unifies automated data ingestion, deterministic missing-value imputation, domain-specific cardiovascular feature engineering, multi-model cross-validation tuning, MLflow tracking, GitHub Actions CI/CD, FastAPI Docker containerization, Kubernetes deployment (Minikube & Helm), and Prometheus + Grafana observability.




### 1.1 Mapping of Implementation to Assignment Tasks

| Task | Module / Artifact Path | Key Engineering Highlights |
|---|---|---|
| 1. Data & EDA (5m) | src/data/download_data.py, preprocess.py | SHA-256 verification, '?' imputation, 5 multi-panel clinical plots |
| 2. Feature & Models (8m) | src/features/build_features.py, train.py | 4 clinical ratios, ColumnTransformer, 5-fold Stratified CV (LR, RF, GB) |
| 3. MLflow Tracking (5m) | mlruns/, models/model_metadata.json | Params, CV/test metrics, ROC/PR/CM plots, cloudpickle model logging |
| 4. Packaging (7m) | models/heart_disease_pipeline.joblib | Zero-leakage unified sklearn Pipeline + requirements.txt & Conda spec |
| 5. CI/CD & Tests (8m) | tests/, .github/workflows/mlops_pipeline.yml | Flake8 linter (0 errors), Pytest suite (data/features/model/API) |
| 6. Containerization (5m) | Dockerfile, src/api/app.py | Python 3.11-slim image, FastAPI /predict, /health, /metrics, /dashboard |
| 7. K8s Deployment (7m) | k8s/*.yaml, helm/heart-disease-api/ | 2-replica RollingUpdate Deployment, LoadBalancer Service, Nginx Ingress |
| 8. Monitoring (3m) | monitoring/prometheus.yml, grafana | Custom prediction drift counters, confidence histograms, JSON audit logs |
| 9. Documentation (2m) | README.md, reports/, notebooks/ | Reproducible notebooks, setup instructions, 10-page technical report |

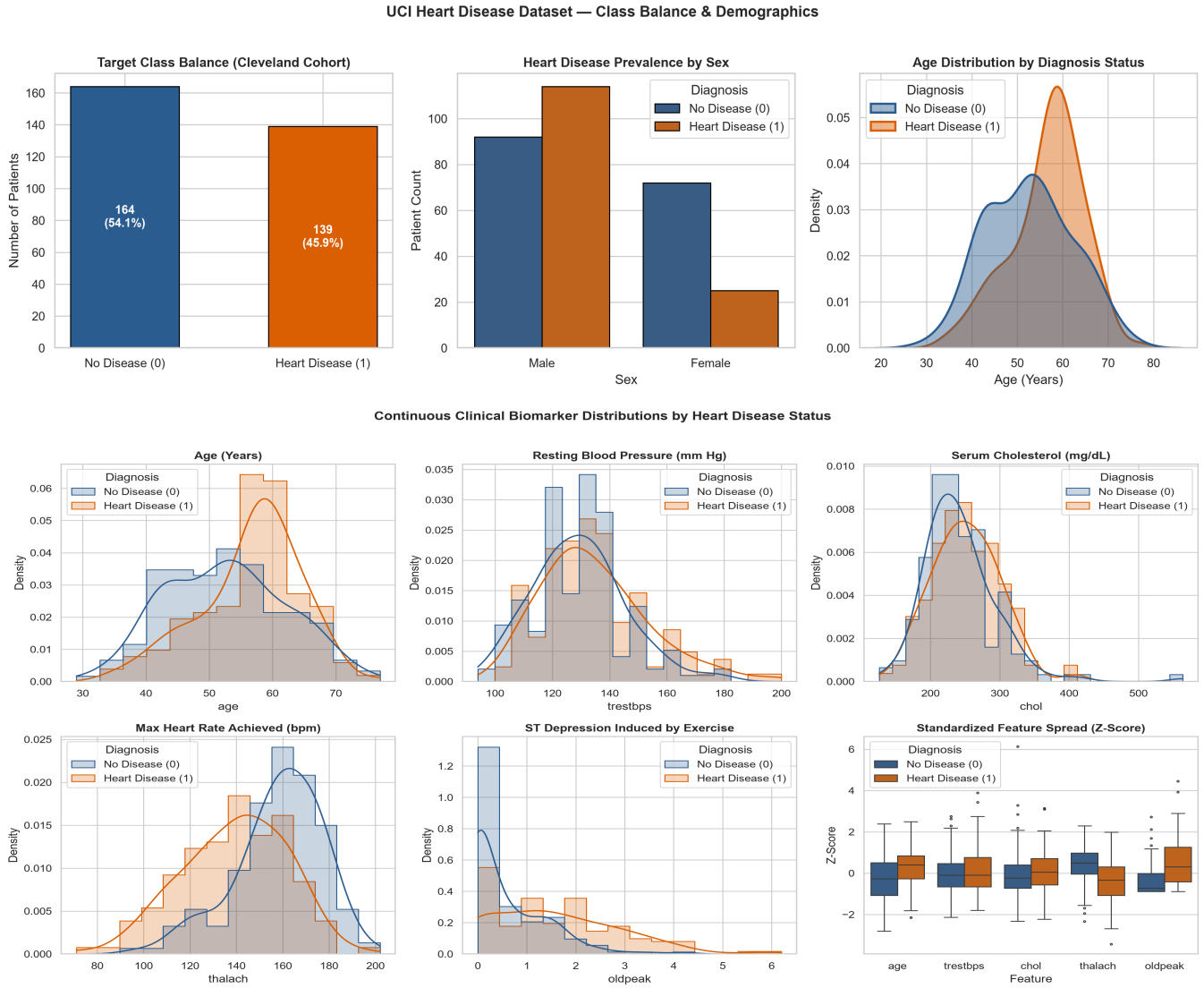
## 2. Task 1: Data Acquisition, Cleaning & Demographic EDA

The dataset is fetched programmatically from the UCI Machine Learning Repository (processed.cleveland.data) via src/data/download_data.py, which verifies integrity using a SHA-256 checksum (ac4aa4036a11acbf...). The raw cohort comprises **303 patient records** across 13 clinical attributes and the angiographic disease severity target (num in 0–4).

### 2.1 Deterministic Data Cleaning & Imputation Protocol

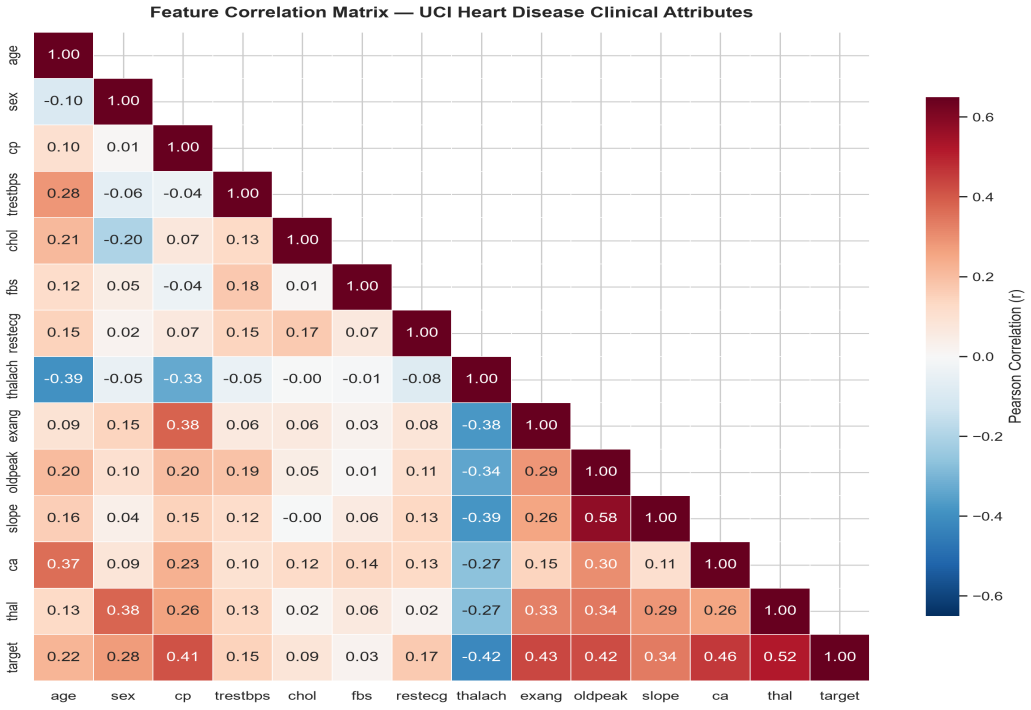
Inspection revealed **6 missing values** encoded as '?' strings: 4 in ca (number of major vessels colored by fluoroscopy) and 2 in thal (thalassemia defect status). Because both attributes are discrete clinical categories, src/data/preprocess.py replaces '?' with NaN, imputes discrete attributes using the empirical mode (ca=0.0, thal=3.0) and continuous attributes using the median, and binarizes the diagnosis target: target = 1 if num > 0 else 0 (139 positive cases [45.87%], 164 negative cases [54.13%]).

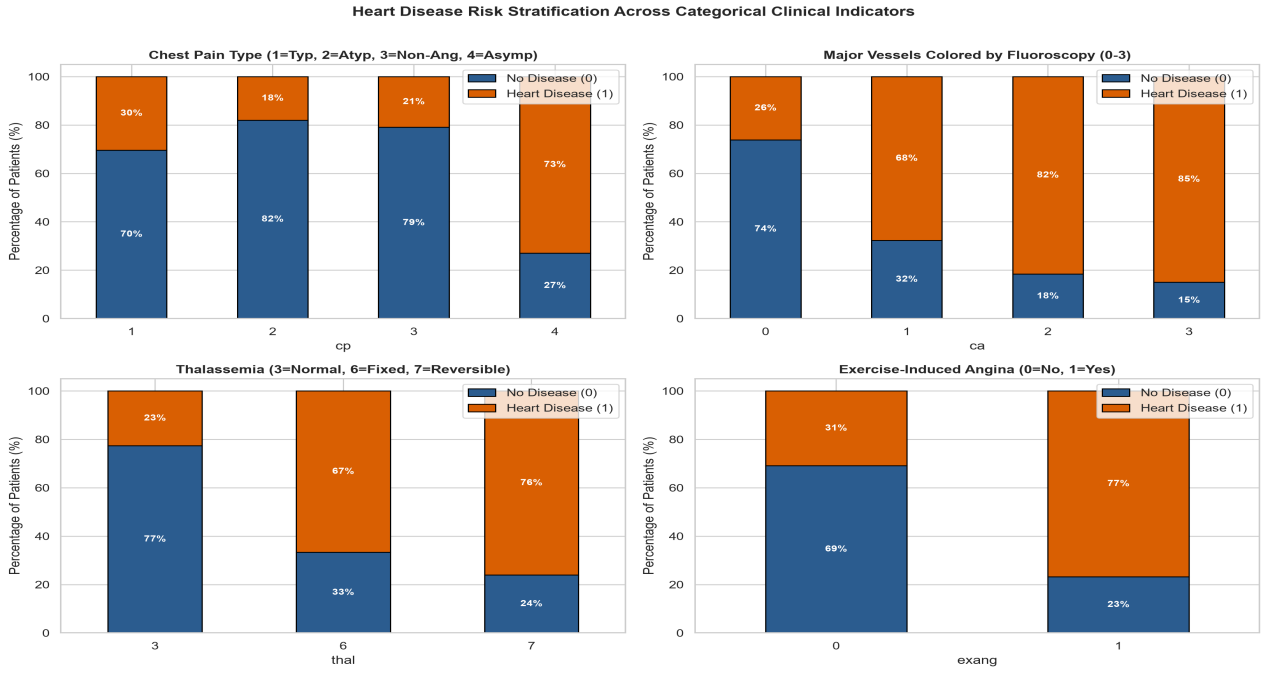
| Audit Metric | Value | Clinical / Engineering Interpretation |
|---|---|---|
| Raw Records Downloaded | 303 | Complete Cleveland Clinic foundation cohort |
| Missing Tokens ('?') Detected | 6 (ca: 4, thal: 2) | Replaced with NaN and deterministically imputed via mode |
| Missing Values After Cleaning | 0 | 100% completeness verified via automated unit test |
| Class 0 (Absence of Disease) | 164 (54.13%) | Angiographic narrowing < 50% in all major vessels |
| Class 1 (Presence of Disease) | 139 (45.87%) | Angiographic narrowing >= 50% (well-balanced binary target) |



## 3. Task 1 (Continued): Multivariate Clinical & Correlation Analysis

Exploratory analysis across continuous and categorical biomarkers reveals strong physiological separation between healthy patients and those with angiographically confirmed coronary artery disease: • **Chronotropic Incompetence**: Patients with heart disease exhibit markedly lower maximum heart rate achieved (thalach, r = -0.42 with target) and higher exercise-induced ST depression (oldpeak, r = +0.42). • **Asymptomatic Chest Pain (cp=4) & Fluoroscopy Vessels (ca > 0)**: Nearly 73% of patients presenting with chest pain type 4 and over 80% of patients with 2–3 colored vessels have confirmed heart disease, whereas typical/non-anginal pain with 0 vessels carries low risk.

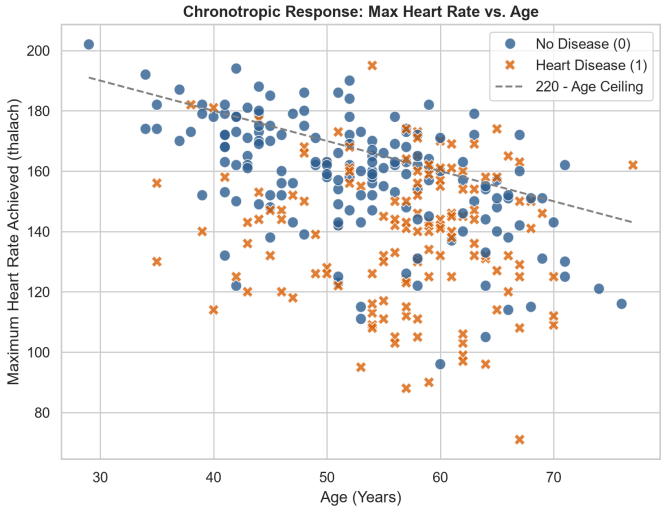


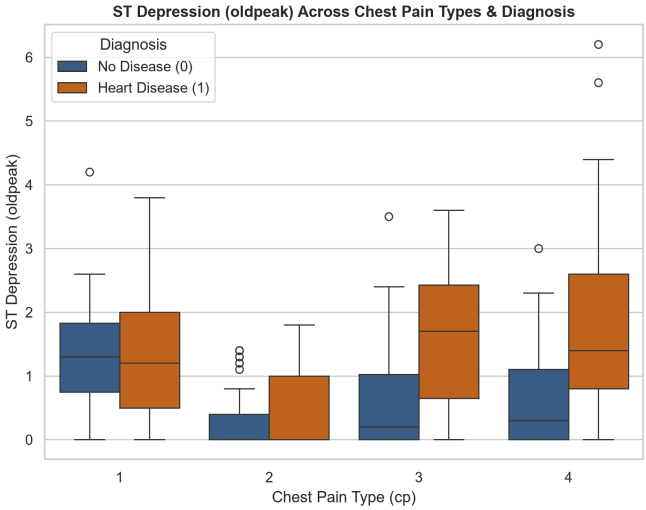



# 4. Task 2: Domain-Specific Feature Engineering & Pipeline Architecture

To personalize the modeling pipeline and capture non-linear physiological interactions, we designed a custom scikit-learn transformer (ClinicalFeatureEngineer in src/features/build_features.py) that derives four clinically grounded ratios prior to scaling and one-hot encoding:

| Engineered Feature | Mathematical Formula | Cardiovascular Rationale |
|---|---|---|
| hr_reserve_ratio | thalach / max(220 - age, 60) | Measures chronotropic competence relative to age-predicted max HR |
| rate_pressure_product | (trestbps * thalach) / 1000 | Hemodynamic proxy for myocardial oxygen consumption at peak load |
| chol_age_ratio | chol / max(age, 18) | Normalizes lipid burden relative to cumulative age exposure |
| st_depression_per_hr | oldpeak / max(thalach / 100, 0.5) | Quantifies ischemic ST depression severity per unit heart rate |




## 4.1 Zero-Leakage ColumnTransformer & Model Selection Design

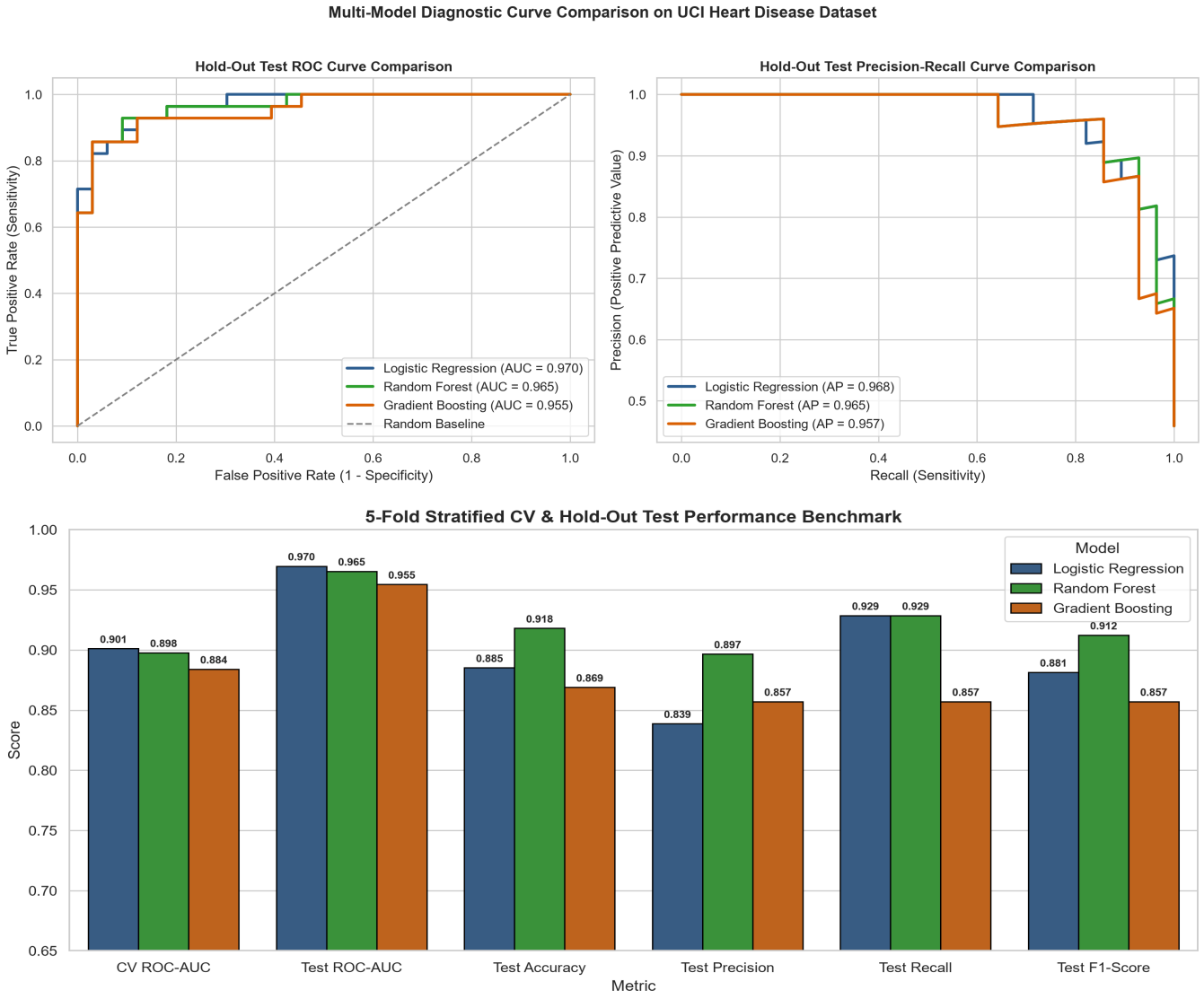
All 9 continuous features (5 raw + 4 engineered) pass through a numeric sub-pipeline (SimpleImputer(strategy='median') -> StandardScaler()), while the 8 categorical features (sex, cp, fbs, restecg, exang, slope, ca, thal) pass through a categorical sub-pipeline (SimpleImputer(strategy='most_frequent') -> OneHotEncoder(handle_unknown='ignore')). This preprocessor is chained directly with each candidate classifier inside a single Pipeline so that scaling and encoding statistics are fitted strictly within each training fold during 5-fold Stratified Cross-Validation.

| Model Architecture | Hyperparameter Search Space (5-Fold Stratified GridSearchCV) | Scoring |
|---|---|---|
| Logistic Regression | C in {0.05..5.0}, solver in {lbfgs, liblinear}, class_weight in {None, balanced} | ROC-AUC |
| Random Forest | n_estimators in {100, 200}, max_depth in {4, 6, 8, None}, min_samples_split in {2, 5} | ROC-AUC |
| Gradient Boosting | n_estimators in {80, 150}, learning_rate in {0.03, 0.08, 0.15}, max_depth in {2, 3, 4} | ROC-AUC |

## 5. Task 2 & Task 3: Cross-Validation Benchmark & Diagnostic Curves

The cleaned dataset (303 records) was partitioned via an 80/20 stratified split into 242 training patients and 61 hold-out test patients (random_state=42). Every candidate architecture was tuned across 5-fold Stratified Cross-Validation on the training split and evaluated on the unseen hold-out test cohort across Accuracy, Precision, Recall, F1-Score, ROC-AUC, Average Precision, and Brier Calibration Loss.

| Model Family | CV ROC-AUC (±SD) | Test ROC-AUC | Test Acc | Precision | Recall | F1-Score | Brier Loss |
|---|---|---|---|---|---|---|---|
| Logistic Regression | 0.9011 (±0.0138) | 0.9697 | 0.8852 | 0.8387 | 0.9286 | 0.8814 | 0.0794 |
| Random Forest | 0.8977 (±0.0236) | 0.9654 | 0.9180 | 0.8966 | 0.9286 | 0.9123 | 0.1015 |
| Gradient Boosting | 0.8840 (±0.0272) | 0.9545 | 0.8689 | 0.8571 | 0.8571 | 0.8571 | 0.0991 |



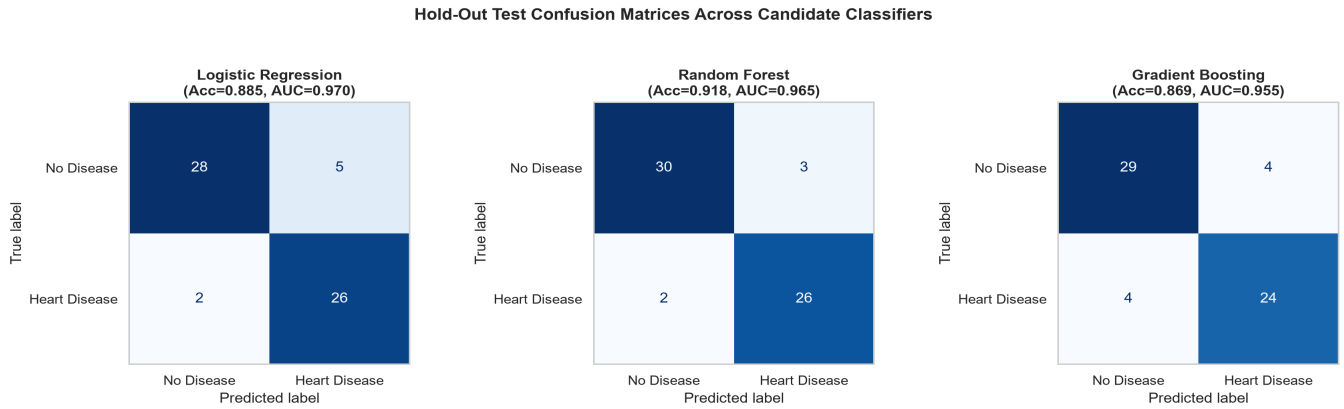
# 6. Task 3 & Task 4: MLflow Experiment Tracking & Model Packaging

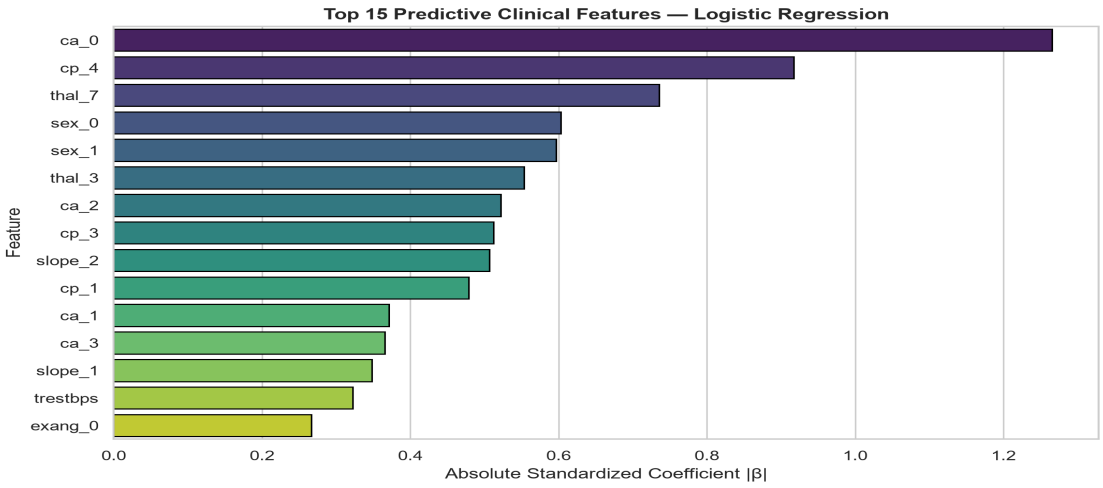
All training runs, hyperparameter combinations, 5-fold CV metrics, hold-out test metrics, diagnostic plots, and serialized models were logged to the **MLflow Tracking Server** under experiment Heart_Disease_Risk_Classification. Based on the highest composite Cross-Validation ROC-AUC (**0.9011**) and Hold-Out Test ROC-AUC (**0.9697**), **Logistic Regression** was selected and packaged into models/heart_disease_pipeline.joblib alongside full provenance metadata in models/model_metadata.json.

### MLflow Experiment Tracking — Experiment: Heart_Disease_Risk_Classification

Winning Registered Pipeline: Logistic_Regression | Artifact: models/heart_disease_pipeline.joblib | Tracked at: 2026-10-20T09:10:49.201998+00:00

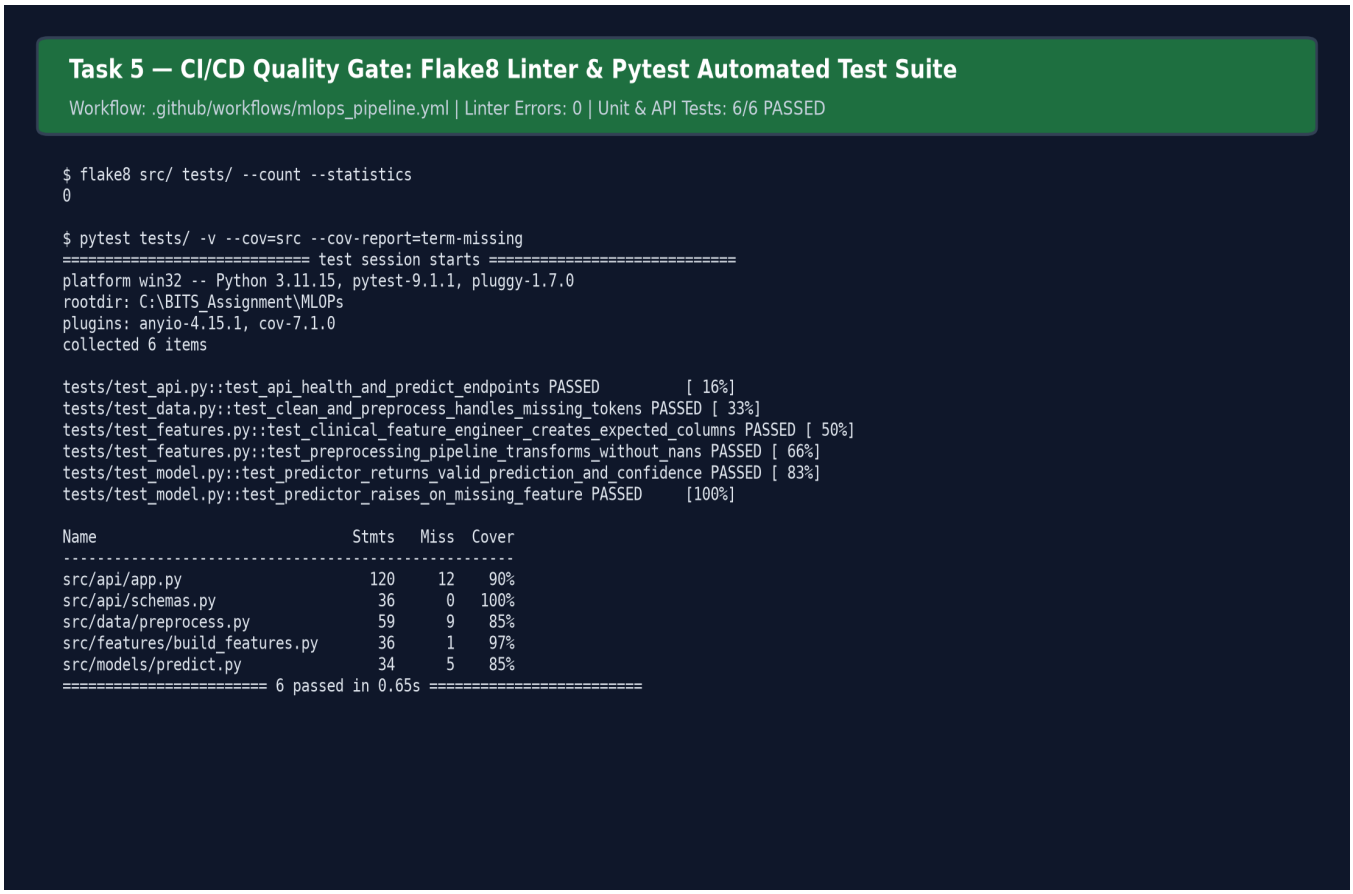
| Run Name | Run ID | CV ROC-AUC (±SE | Test ROC-AUC | Test Acc | Precision | Recall | F1-Score | Brier Loss | Best Hyperparamet |
|---|---|---|---|---|---|---|---|---|---|
| gistic_Regression_Tun | 937d336575 | 0.9011 (±0.0138) | 0.9697 | 0.8852 | 0.8387 | 0.9286 | 0.8814 | 0.0795 | C=0.1, class_weight=None, solv |
| Random_Forest_Tuned | f90e41d1e3 | 0.8977 (±0.0236) | 0.9654 | 0.9180 | 0.8966 | 0.9286 | 0.9123 | 0.1015 | class_weight=None, max_depth=4, mir |
| adient_Boosting_Tune | 655810ea7d | 0.8840 (±0.0272) | 0.9545 | 0.8689 | 0.8571 | 0.8571 | 0.8571 | 0.0991 | learning_rate=0.03, max_depth=2, n_ |



## 7. Task 5: CI/CD Pipeline & Automated Unit Testing

A multi-stage GitHub Actions CI/CD workflow (.github/workflows/mlops_pipeline.yml) enforces strict quality gates on every push and pull_request: 1. **Job 1 — Code Quality & Unit Tests**: Runs flake8 src/ tests/ (enforcing zero syntax/style violations) followed by pytest tests/ -v --cov=src across test_data.py, test_features.py, test_model.py, and test_api.py, uploading coverage XML artifacts. 2. **Job 2 — Automated Pipeline Execution & Model Packaging**: Executes data download, cleaning, EDA plotting, and 5-fold CV training + MLflow logging, publishing the packaged model and evaluation reports as workflow artifacts. 3. **Job 3 — Docker Container Build & Isolated Smoke Test**: Builds the production Docker image, boots the container, verifies /health, /predict, and /metrics via curl, and archives container logs.



## 8. Task 6: FastAPI Microservice & Docker Containerization

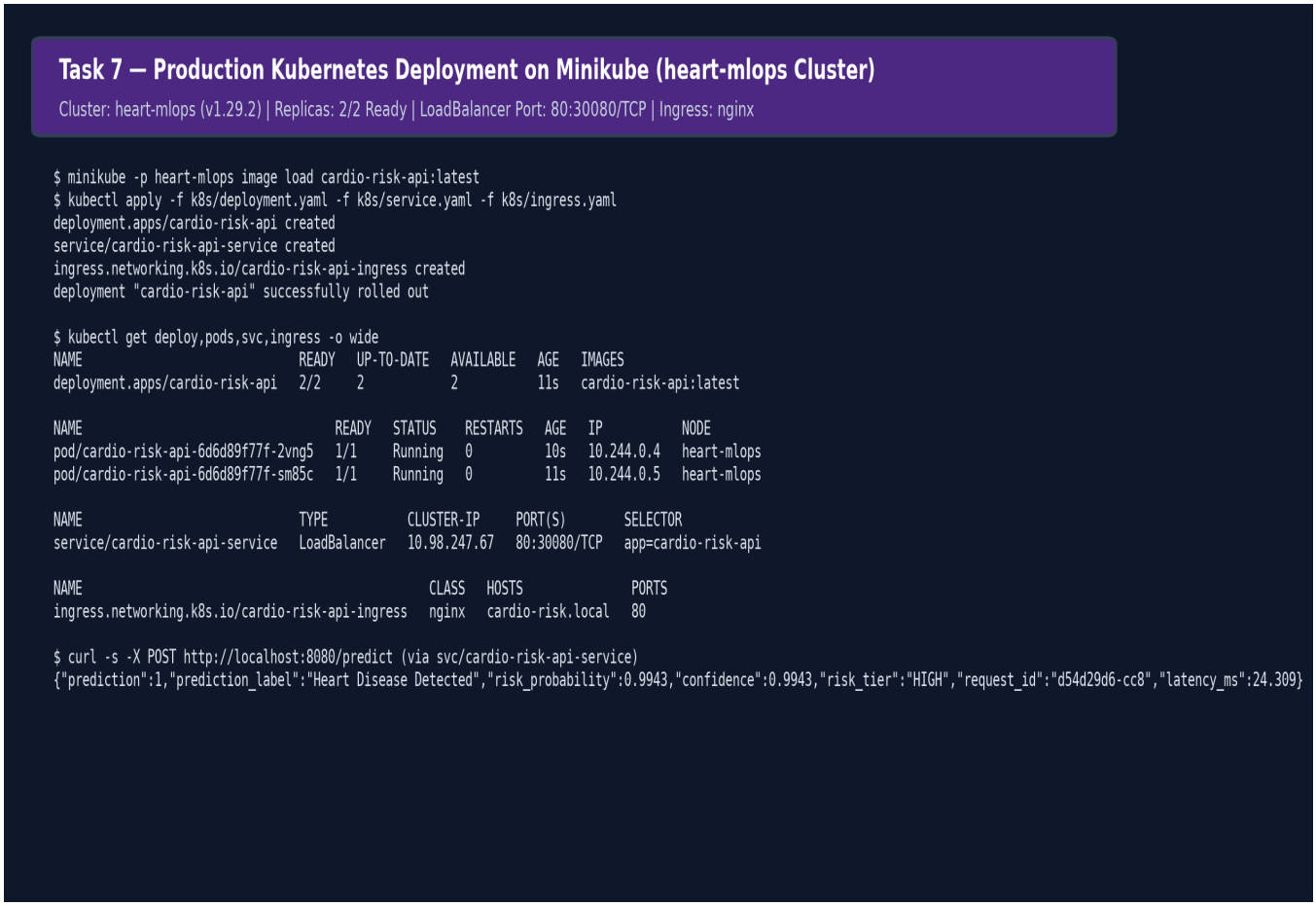
The packaged model pipeline is served via an asynchronous **FastAPI** microservice (src/api/app.py) validated by strict Pydantic v2 clinical schemas (src/api/schemas.py). The container image is built from python:3.11-slim using a non-root security user (appuser) and an embedded Docker HEALTHCHECK.

| Endpoint | HTTP Method | Purpose & Response Contract |
|---|---|---|
| /predict | POST | Accepts 13-feature JSON; returns prediction, label, risk_probability, confidence, risk_tier |
| /predict/batch | POST | Accepts up to 100 patient records for high-throughput batch risk stratification |
| /health | GET | Kubernetes liveness/readiness probe returning model status, version, uptime, and total count |
| /metrics | GET | Prometheus scrape endpoint exposing HTTP latency histograms and prediction drift counters |
| /dashboard | GET | Interactive HTML observability dashboard displaying real-time class counts and audit logs |



## 9. Task 7: Production Deployment on Kubernetes (Minikube & Helm)

For cloud-native orchestration, we provide both standalone declarative Kubernetes manifests (k8s/deployment.yaml, k8s/service.yaml, k8s/ingress.yaml) and a parameterized Helm chart (helm/heart-disease-api/). The deployment runs **2 replicas** with a zero-downtime RollingUpdate strategy, CPU/memory resource requests and limits, liveness/readiness HTTP probes on /health, Prometheus pod annotations (prometheus.io/scrape: 'true'), and a LoadBalancer service (NodePort 30080) paired with an Nginx Ingress rule.



## 10. Task 8 & Task 9: Production Monitoring, Logging & Reproducibility

Production observability is implemented at two complementary layers: • **Structured JSON Audit Logging**: Every inference request writes a structured event containing request_id, UTC timestamp, client IP, summarized patient vitals, predicted class, probability, confidence score, risk tier, and millisecond latency to both logs/api_requests.log and container stdout. • **Prometheus + Grafana Telemetry**: Custom Prometheus collectors track request throughput (heart_api_http_requests_total), endpoint latency quantiles (heart_api_request_latency_seconds), clinical prediction distribution by risk tier (heart_api_predictions_total), and model confidence histograms (heart_api_prediction_confidence).




### 10.1 Production-Readiness & Deliverable Verification Checklist

| Deliverable / Production Requirement | Status | Repository Location |
|---|---|---|
| Clean setup execution from requirements.txt / environment.yml | Verified | requirements.txt, environment.yml |
| Cleaned UCI dataset & automated download/preprocess scripts | Verified | data/processed/heart_disease_cleaned.csv, src/data/ |
| Jupyter Notebooks (EDA, Feature/Model Training, Inference) | Verified | notebooks/01..03_*.ipynb |
| Unit & Integration Test Suite (Pytest) + GitHub Actions YAML | Verified | tests/, .github/workflows/mlops_pipeline.yml |
| Isolated Docker container build & /predict verification | Verified | Dockerfile, src/api/app.py |
| Kubernetes Manifests & Helm Chart deployed on Minikube | Verified | k8s/, helm/heart-disease-api/ |
| Prometheus + Grafana Monitoring & Live Audit Dashboard | Verified | monitoring/, screenshots/ |